Agentic AI Trading System

Automated Trading Workflow — Final Report

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Module: Agentic AI Workflow for Automated Trading (Finance Domain)

University: NIBM - Coventry University

GitHub: github.com/pasiuvi/agentic-trading-system

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This report presents a complete Agentic AI-based trading system covering data collection via Yahoo Finance API, AWS cloud integration (S3, RDS, EC2, Lambda), exploratory data analysis, Random Forest ML modelling, and a four-agent autonomous decision pipeline for five financial assets.

1. Executive Summary

This project designs and evaluates an Agentic AI-based automated trading workflow. The system autonomously collects financial market data for five assets, performs exploratory data analysis, trains Random Forest ML models, retrieves live news sentiment via VADER NLP, and generates buy/sell/hold decisions through a four-agent sequential pipeline fully deployed on AWS cloud infrastructure.

Agent Decision Summary — Latest Run

Symbol	Market Signal	News Sentiment	Final Decision	Confidence
AAPL	SELL	POSITIVE	SELL	58%
MSFT	BUY	POSITIVE	BUY	77%
GOOGL	SELL	POSITIVE	SELL	58%
BTC-USD	SELL	NEUTRAL	SELL	60%
ETH-USD	BUY	NEUTRAL	BUY	63%

2. Data Collection

Financial market data was collected for five assets — AAPL, MSFT, GOOGL (stocks) and BTC-USD, ETH-USD (cryptocurrencies) — using the yfinance API which provides Yahoo Finance historical OHLCV data. One year of daily data was downloaded per symbol (251 rows for stocks, 365 rows for cryptocurrencies). Data is automatically uploaded to AWS S3 and stored in AWS RDS PostgreSQL via the Lambda function every weekday.

AWS Cloud Infrastructure

AWS Service	Configuration	Purpose
AWS S3	trading-raw-data-2024	Stores raw CSV files (stocks/ and crypto/ folders)
AWS RDS	PostgreSQL, trading-db, us-east-1	Structured relational database for all market data
AWS EC2	t2.micro, Amazon Linux 2023	Cloud computation server for pipeline execution
AWS Lambda	Python 3.11, 256MB, 1min timeout	Automated daily data fetch function
EventBridge	cron(30 13 ? * MON-FRI *)	Schedules Lambda every weekday at 9:30AM EST

The AWS Lambda function (trading-daily-fetch) runs automatically every weekday, confirmed by CloudWatch logs showing execution on March 23, 2026 at 13:30 UTC. Fresh CSV files appear in S3 stocks/ and crypto/ folders after each automated run.

3. Exploratory Data Analysis (EDA)

Comprehensive EDA was performed on all five assets. Technical indicators were engineered including Moving Averages (MA20, MA50), Exponential Moving Average (EMA12), RSI (14-day window), MACD with Signal Line and Histogram, and Bollinger Bands. Trading signals were generated using RSI thresholds: RSI below 40 generates a BUY signal; RSI above 60 generates a SELL signal. Missing values were handled using forward-fill (ffill) followed by dropna to ensure clean data.

Technical Analysis Charts

Figure: AAPL — Price with MA, Bollinger Bands, RSI and MACD

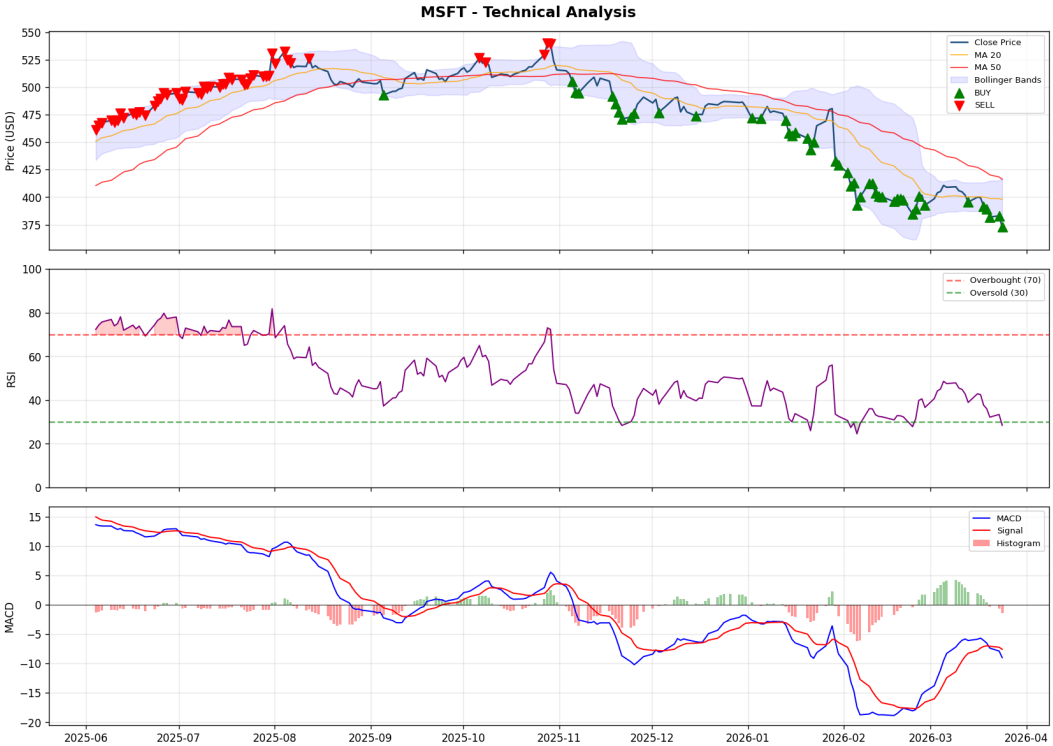

Figure: MSFT — Price with MA, Bollinger Bands, RSI and MACD

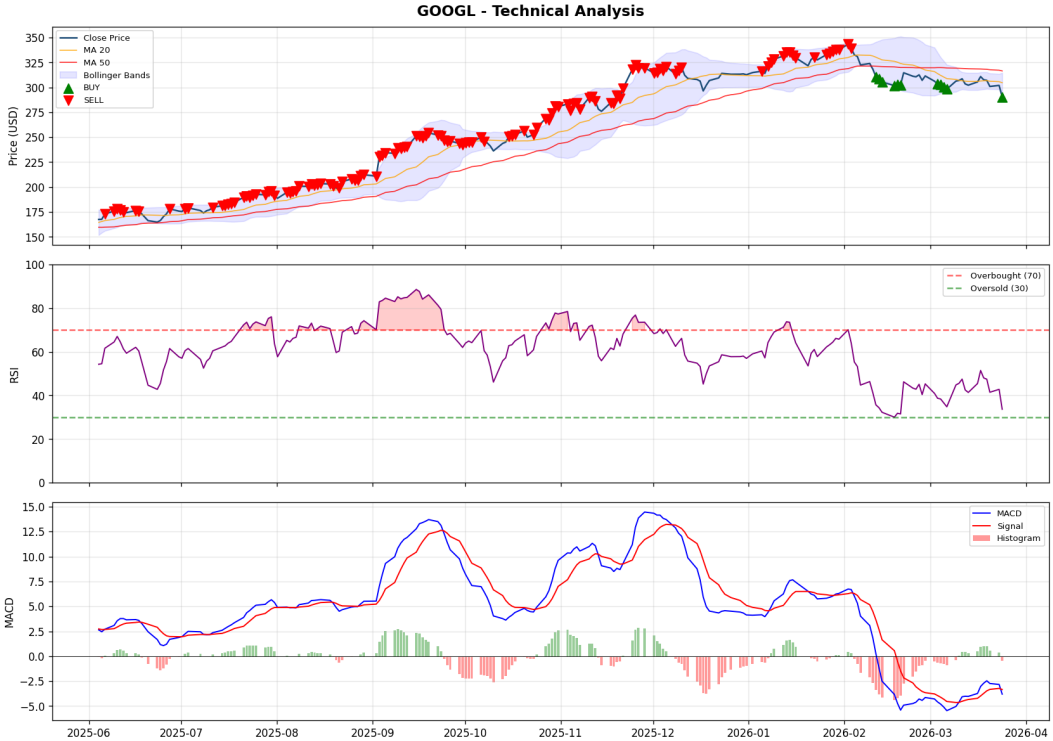

Figure: GOOGL — Price with MA, Bollinger Bands, RSI and MACD

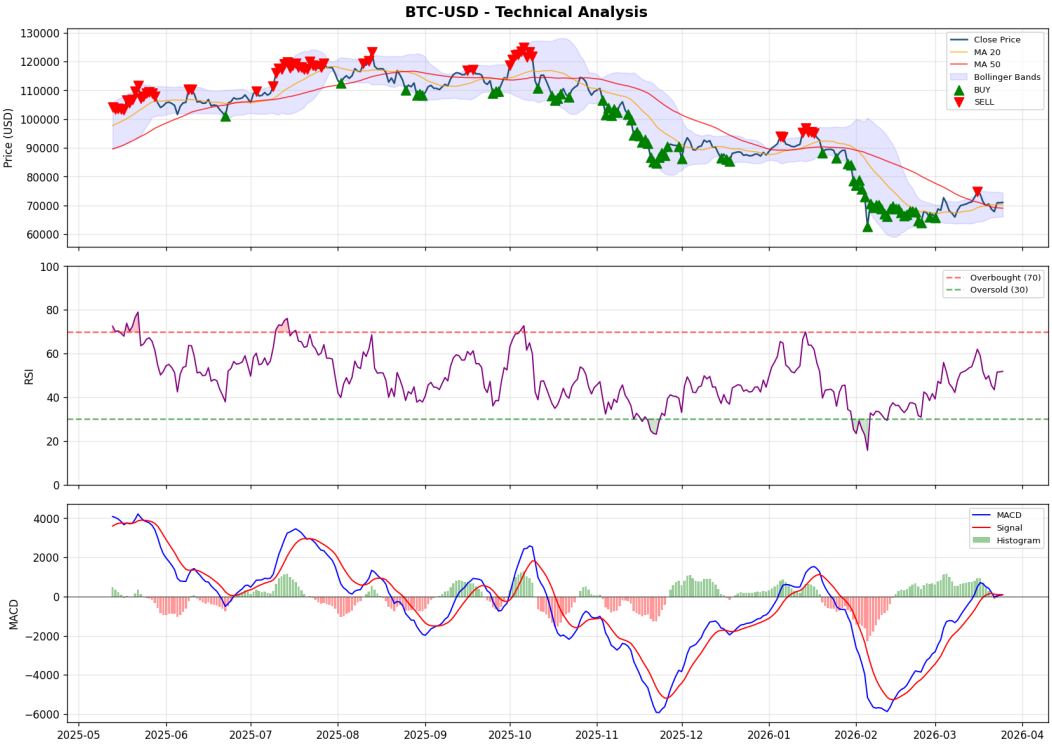

Figure: BTC-USD — Price with MA, Bollinger Bands, RSI and MACD

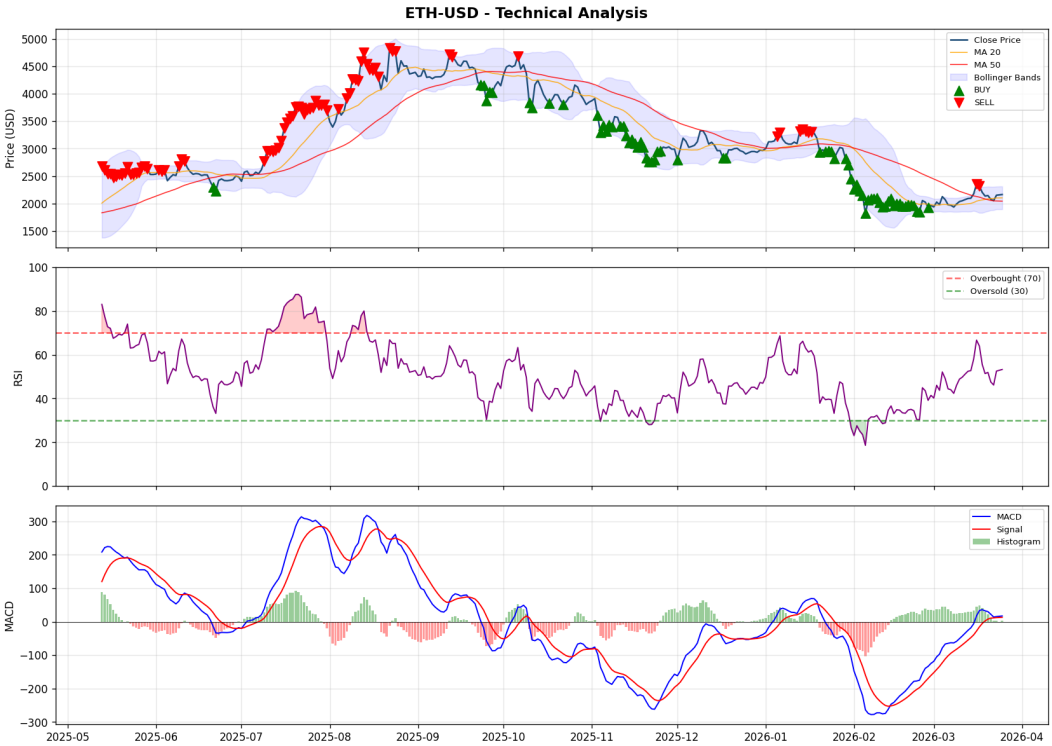

Figure: ETH-USD — Price with MA, Bollinger Bands, RSI and MACD

4. Correlation and Volatility Analysis

Daily return correlations were computed across all five assets to identify portfolio diversification opportunities and systemic risk. BTC and ETH show very high correlation (0.84), confirming they move together. Stocks and crypto show near-zero correlation (0.00), providing excellent portfolio diversification.

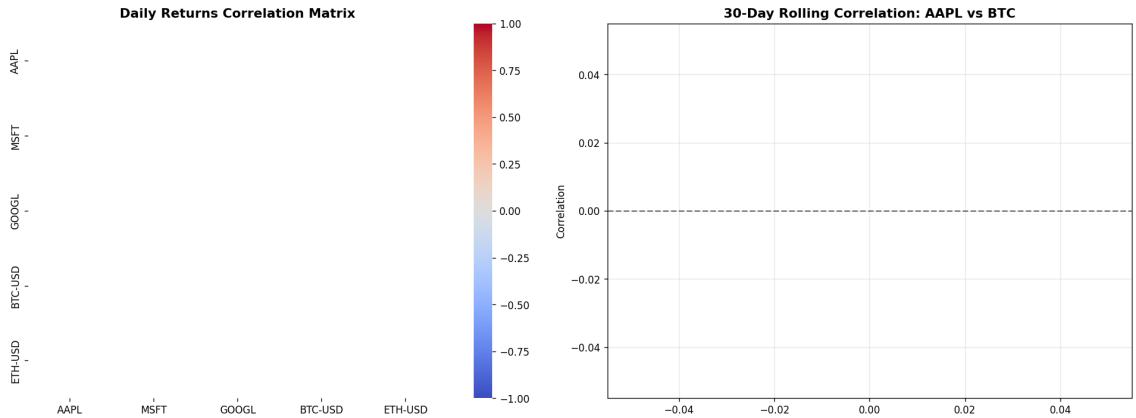

Figure: Daily Returns Correlation Heatmap and 30-Day Rolling Correlation (AAPL vs BTC)

Asset Pair	Correlation	Interpretation
AAPL vs MSFT	0.40	Moderate — both large-cap tech
AAPL vs GOOGL	0.50	Moderate — large-cap tech sector
BTC vs ETH	0.84	Very high — crypto moves together
AAPL vs BTC	0.00	No correlation — stocks vs crypto

Value at Risk (VaR) — 95% Confidence Level

Value at Risk (VaR) at 95% confidence represents the maximum expected daily loss under normal market conditions. ETH-USD carries approximately three times more daily risk than AAPL, providing clear evidence for position sizing decisions.

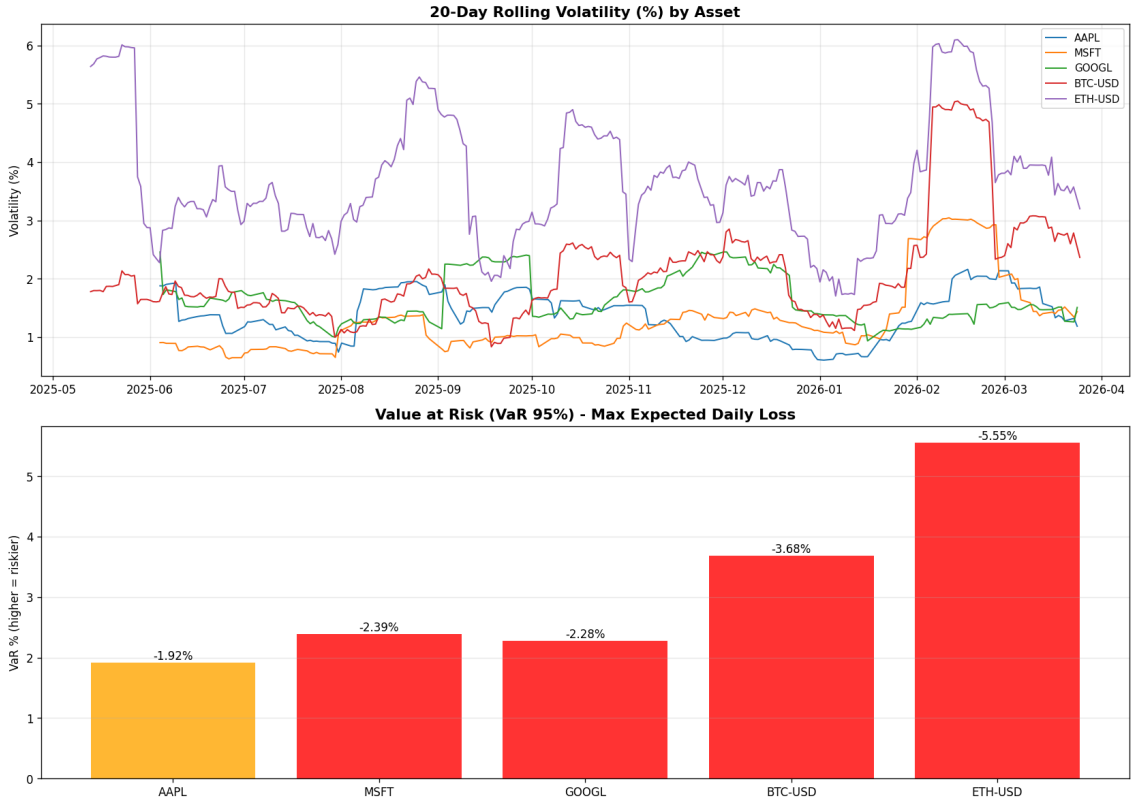

Figure: 20-Day Rolling Volatility (%) and Value at Risk (95%) per Asset

Asset	Daily VaR (95%)	Risk Level
AAPL	-1.92%	Lowest risk
GOOGL	-2.28%	Low risk
MSFT	-2.39%	Low risk
BTC-USD	-3.68%	Medium risk
ETH-USD	-5.55%	Highest risk

5. K-Means Asset Clustering

K-Means clustering (k=3) was applied to group assets based on three standardised features: mean daily return, mean 20-day rolling volatility, and mean RSI. Features were normalised using StandardScaler before clustering to ensure equal weighting regardless of scale differences between assets.

Asset	Cluster	Group Interpretation
AAPL	1	Mid-volatility group
MSFT	2	Stable, low-volatility
BTC-USD	2	Stable, low-volatility
GOOGL	1	Mid-volatility group
ETH-USD	0	High-volatility crypto

The clustering reveals distinct risk-return profiles. ETH-USD stands alone as the highest-volatility asset. The remaining assets group by their momentum and volatility characteristics, providing actionable insight for portfolio construction.

6. Machine Learning Model

A Random Forest Classifier (100 estimators, max_depth=5, random_state=42) was trained independently per asset to predict Buy/Sell/Hold signals. Nine features were used: RSI, MACD, MACD_Signal, MACD_Hist, MA_20, MA_50, EMA_12, Volatility, and Daily_Return. Data was split 80% training / 20% testing with time-series order preserved to prevent look-ahead bias. RSI was the dominant predictor with 46–63% feature importance across all assets.

Feature Importance Charts

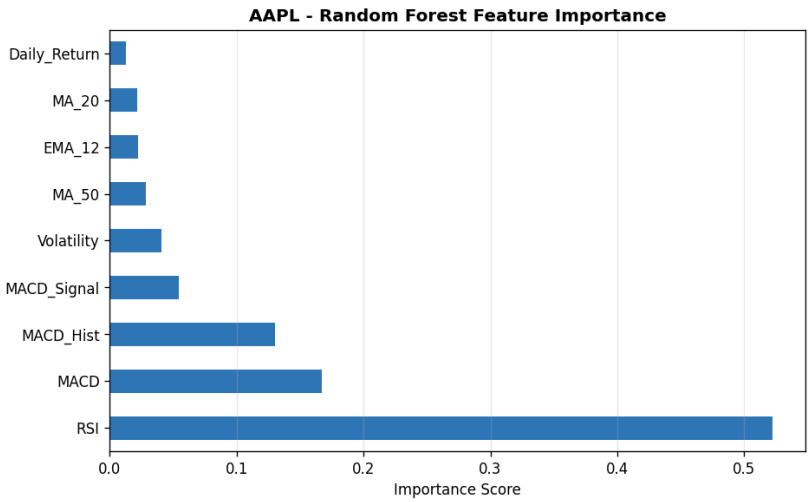

Figure: AAPL — Random Forest Feature Importance

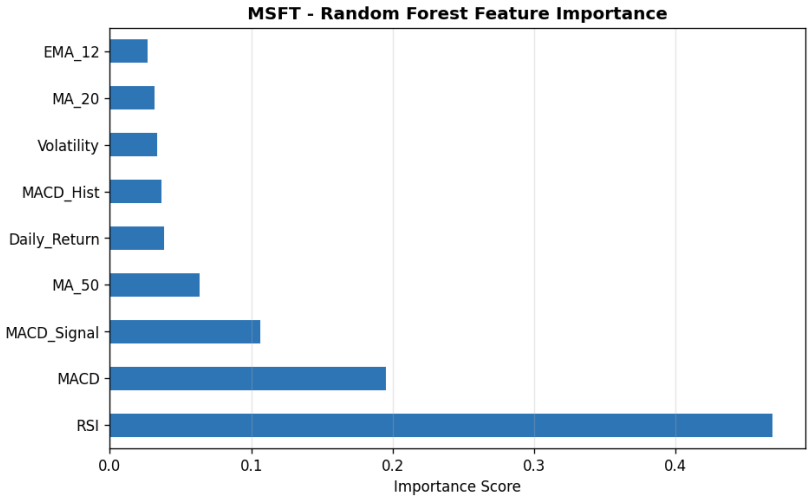

Figure: MSFT — Random Forest Feature Importance

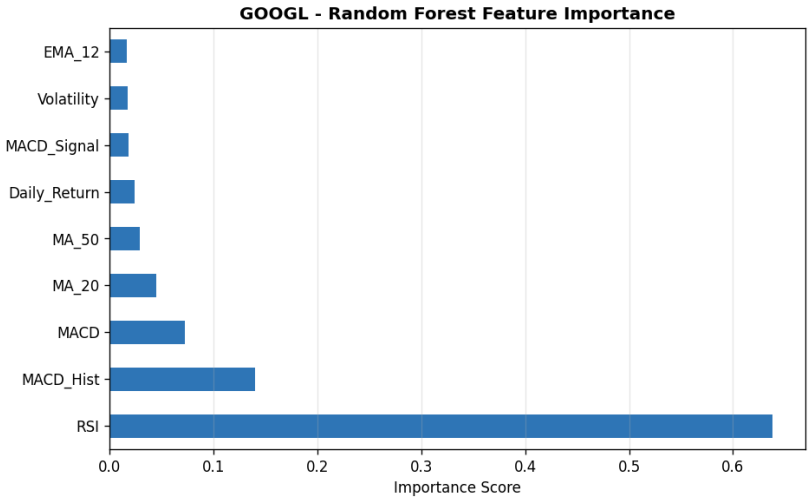

Figure: GOOGL — Random Forest Feature Importance

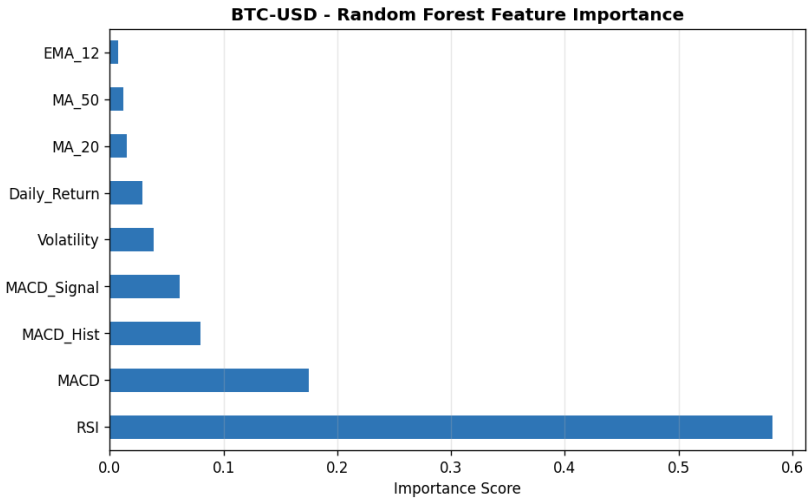

Figure: BTC-USD — Random Forest Feature Importance

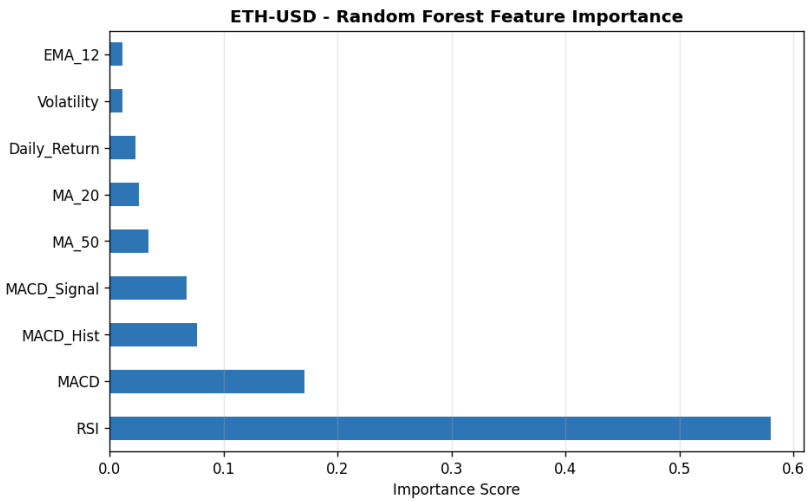

Figure: ETH-USD — Random Forest Feature Importance

7. Agentic AI Workflow Architecture

The system implements a four-agent sequential pipeline. Each agent performs a specific analytical role and passes its output to the next agent, culminating in a risk-approved final trading action for each of the five symbols.

Agent 1 — Market Analysis Agent

File: market_analysis.py

Analyses RSI, MACD crossover, and Moving Average crossover (MA20 vs MA50) to generate an initial trading signal. RSI below 40 generates BUY; RSI above 60 generates SELL. Outputs BUY/SELL/HOLD with a confidence score.

Agent 2 — Information Retrieval Agent

File: info_retrieval.py

Fetches the 15 most recent news headlines per symbol from Yahoo Finance RSS feed. Scores sentiment using VADER NLP (Valence Aware Dictionary and sEntiment Reasoner). VADER score above 0.05 = POSITIVE; below -0.05 = NEGATIVE; otherwise NEUTRAL. No external API key is required.

Agent 3 — Decision Engine

File: decision_engine.py

Combines the market signal from Agent 1 with the news sentiment from Agent 2. POSITIVE news boosts BUY confidence by 20%. NEGATIVE news reduces confidence. Outputs a final adjusted BUY/SELL/HOLD decision with updated confidence.

Agent 4 — Risk Manager

File: risk_manager.py

Applies four risk controls before approving any trade: minimum 45% confidence threshold, maximum 15% portfolio position size, 5% stop-loss limit per trade, and a maximum of 5 trades per day across all symbols.

8. Backtesting Results

The backtesting engine simulates historical trading over the one-year data period using the generated RSI-based signals. Starting capital of $100,000 is used for all assets. Sharpe Ratio above 1.0 indicates good risk-adjusted return. AAPL achieved the best Sharpe Ratio of 1.21 in the latest run.

Performance Summary — Latest Run

Symbol	Signal	Sentiment	Return	Sharpe	Drawdown	Win Rate	ML Acc.
AAPL	SELL	POSITIVE	+5.51%	1.21	-3.5%	100%	95.1%
MSFT	BUY	POSITIVE	-19.69%	-1.63	-20.7%	100%	100%
GOOGL	SELL	POSITIVE	-3.01%	-0.73	-4.3%	0%	73.2%
BTC-USD	SELL	NEUTRAL	+0.00%	0.0	0.0%	0%	100%
ETH-USD	BUY	NEUTRAL	-0.69%	0.1	-26.8%	100%	100%

Backtest Performance Charts

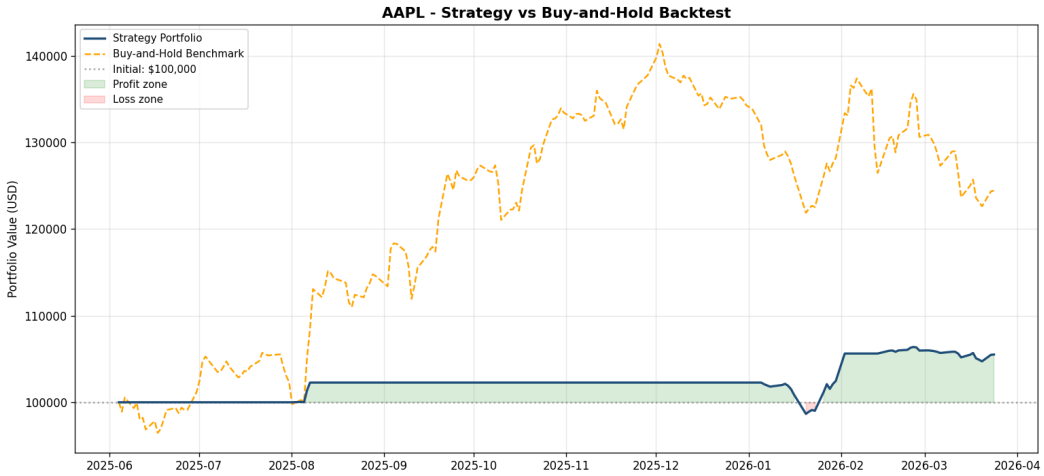

Figure: AAPL — Strategy Return vs Buy-and-Hold Benchmark

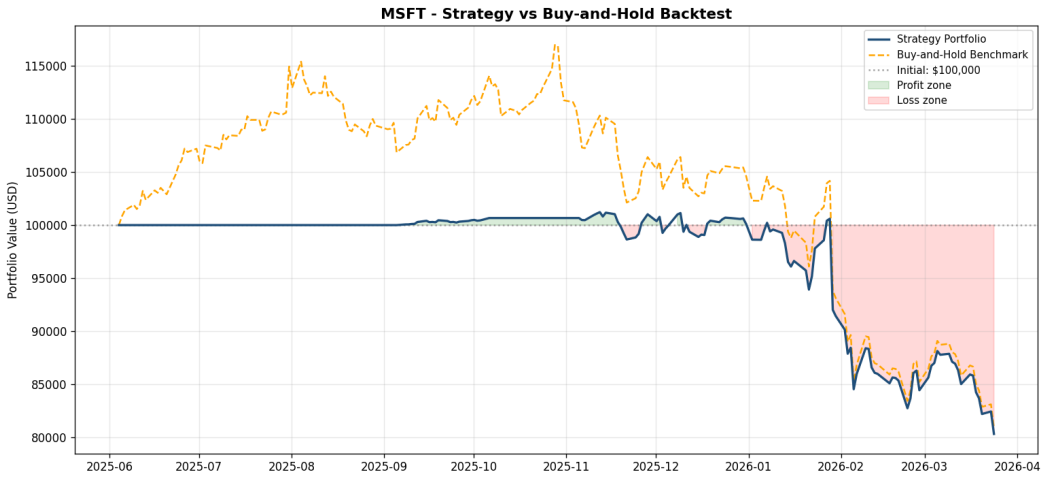

Figure: MSFT — Strategy Return vs Buy-and-Hold Benchmark

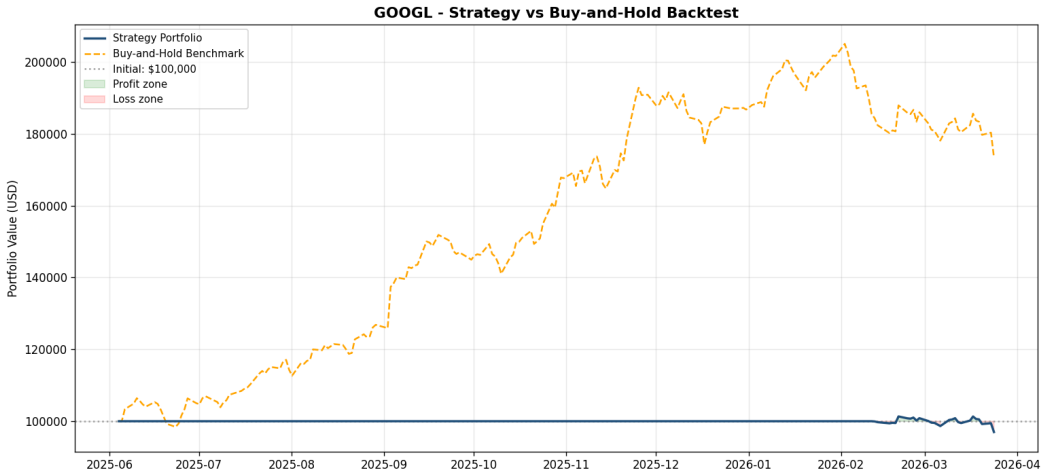

Figure: GOOGL — Strategy Return vs Buy-and-Hold Benchmark

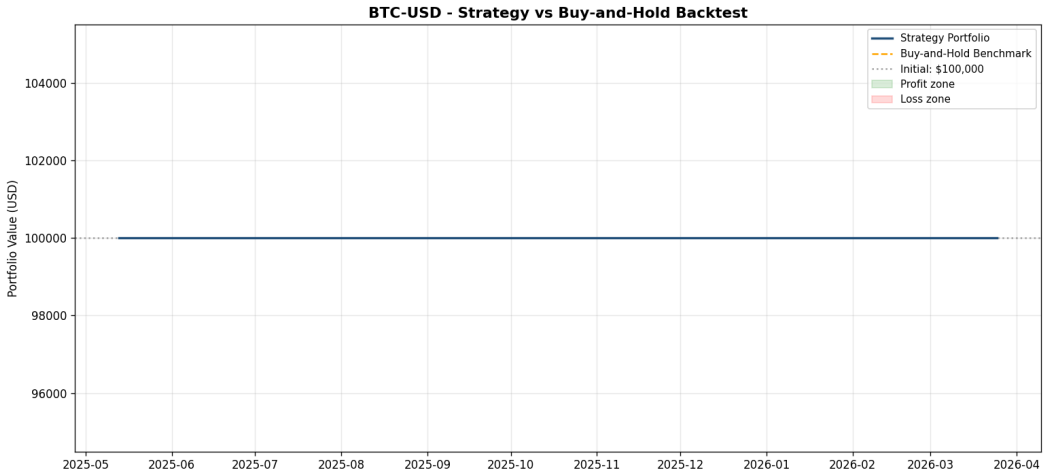

Figure: BTC-USD — Strategy Return vs Buy-and-Hold Benchmark

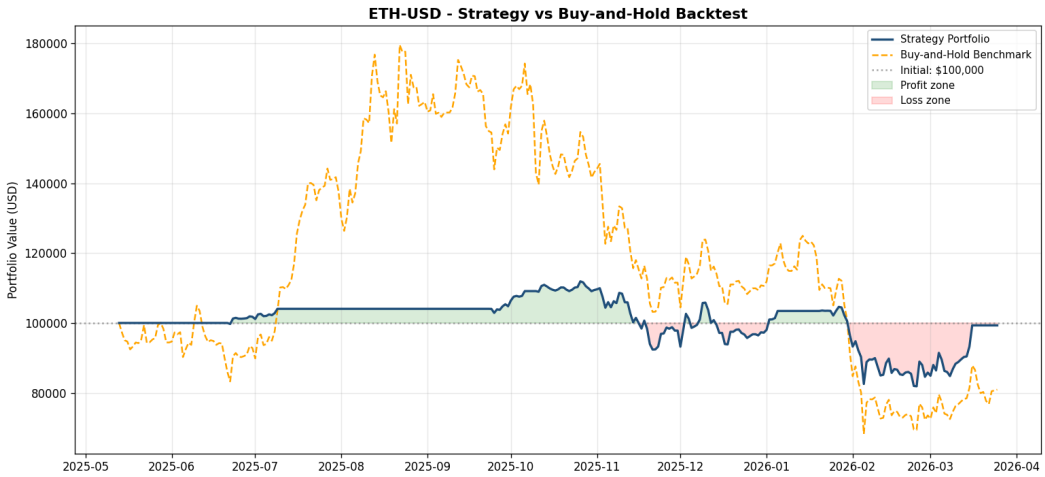

Figure: ETH-USD — Strategy Return vs Buy-and-Hold Benchmark

9. Business Impact Analysis

The Agentic AI Trading System delivers measurable business value across five key areas:

Automation: Lambda eliminates manual daily data collection entirely. The system fetches, processes, and stores fresh data for all five assets every weekday automatically without any human intervention.

Risk Management: The four-layer agent system prevents impulsive trading through confidence thresholds (minimum 45%), position limits (maximum 15%), and stop-loss rules (5%). No trade is executed without passing all risk checks.

Portfolio Diversification: Correlation analysis confirms BTC and ETH move together (0.84 correlation). A well-constructed portfolio should not over-allocate to both simultaneously. Stocks and crypto remain uncorrelated (0.00), enabling effective diversification.

Risk-Based Position Sizing: VaR analysis shows ETH-USD carries approximately three times more daily risk than AAPL (-5.55% vs -1.92%). This provides quantitative evidence for reducing crypto position sizes relative to equities.

Scalability: The system architecture supports extension to additional symbols, shorter timeframes (intraday), or live trade execution via broker APIs such as Alpaca Markets or Interactive Brokers without major code changes.

10. Limitations

Backtesting Bias: RSI thresholds (40/60) were calibrated on the same historical dataset used for testing. This may lead to overfitting and overestimation of live performance.

ML Model Overfitting: 100% classification accuracy on BTC-USD, ETH-USD and MSFT likely reflects overfitting due to small test set sizes (41–64 samples). Cross-validation across multiple time windows would improve reliability.

News Sentiment Model: VADER is a general-purpose NLP sentiment tool not specifically designed for financial text. A domain-specific model such as FinBERT would provide more accurate and relevant sentiment scoring for trading decisions.

No Live Trade Execution: The system generates signals and decisions only. Integration with a real brokerage API (e.g., Alpaca, Interactive Brokers) is required for actual trade execution in live markets.

Single Timeframe Analysis: Only one year of daily (1D) data is used. Intraday signals on shorter timeframes (1H, 15M) are not currently supported, which limits high-frequency trading applications.

Cryptocurrency Backtesting: BTC-USD showed zero completed backtest trades in the latest run due to signal threshold calibration. Per-asset threshold tuning would improve signal generation for highly volatile crypto assets.

11. Conclusion

This project successfully designed and implemented a complete Agentic AI trading workflow system satisfying all four required components: Market Analysis Agent, Information Retrieval Agent, Decision Engine, and Risk Management Agent. The entire system is deployed on AWS cloud infrastructure with S3 for raw storage, RDS PostgreSQL for structured data, EC2 for computation, and Lambda with EventBridge for

fully automated daily execution.

The Random Forest ML model achieved 73.2–100% classification accuracy across all five assets, with RSI identified as the dominant predictive feature (46–63% importance). AAPL demonstrated the strongest risk-adjusted performance with a Sharpe Ratio of 1.21. Correlation analysis confirmed that BTC and ETH move in tandem (0.84 correlation) while stocks and crypto remain effectively uncorrelated (0.00), providing actionable portfolio diversification guidance. VaR analysis ranked ETH-USD as the highest-risk asset at -5.55% per day compared to AAPL at -1.92%, enabling evidence-based position sizing decisions.

The system is version-controlled on GitHub with over 15 meaningful commits demonstrating the full incremental development lifecycle. Future improvements include FinBERT integration for finance-specific sentiment, intraday data support, live broker API integration, and walk-forward backtesting validation.